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Spectra I

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Last time. Some axiomatics of higher categories,  
(commutative) monoids

Today. More on commutative monoids, Spectra

Recall  $\mathcal{C}$  an  $\infty$ -category with finite products. A monoid object of  $\mathcal{C}$  is a functor  $X: \Delta^{\text{op}} \rightarrow \mathcal{C}$  satisfying:

(1)  $X_0 \xrightarrow{\sim} 1_{\mathcal{C}}$

(2) For each  $n \geq 2$  and  $i \in [n]$ , the square

$$\begin{array}{ccc} X([n]) & \longrightarrow & X(\{i < \dots < n\}) \\ \downarrow & & \downarrow \\ X(\{0 < \dots < i\}) & \longrightarrow & X(\{i\}) \simeq 1_{\mathcal{C}} \end{array}$$

is a pullback.

A commutative monoid in  $\mathcal{C}$  is a functor  $X: \text{Set}_*^{\text{fin}} \longrightarrow \mathcal{C}$  such that:

(1)  $X(*) \xrightarrow{\sim} 1_{\mathcal{C}}$

(2) For all finite pointed sets  $S$  and  $T$ , the functor  $X$  sends the square

$$\begin{array}{ccc} S \vee T & \xrightarrow{\text{collapse}} & T \\ \text{collapse} \downarrow & & \downarrow \\ S & \longrightarrow & * \end{array}$$

to a pullback square

In particular,  $X(S \vee T) \xrightarrow{\sim} X(S) \times X(T)$

## Comparing monoids & commutative monoids

**Definition.** Let  $I \in \Delta$ . The set of cuts of  $I$  is the quotient of the set of disjoint decompositions  $I = S \sqcup T$  such that for all  $s \in S$  and  $t \in T$ , we have  $s < t$  where we identify  $I \sqcup \emptyset$  and  $\emptyset \sqcup I$ .

The assignment  $I \mapsto \text{Cut}(I)$  defines a functor

$$\text{Cut} : \Delta^{\text{op}} \longrightarrow \text{Set}_*^{\text{fin}}$$

with functoriality given by sending  $\phi : I \longrightarrow J$  to the map

$$\begin{aligned} \phi^* : \text{Cut}(J) &\longrightarrow \text{Cut}(I) \\ J = S \sqcup T &\longmapsto I = \phi^{-1}(S) \sqcup \phi^{-1}(T). \end{aligned}$$

Check  $\text{Cut}([n]) \cong \{1, \dots, n\}_+$

**Exercise.** The functor  $\text{Cut}$  is isomorphic to the pointed simplicial set  $\Delta^1 / \partial \Delta^1$ . (This is the 'simplicial circle'.)

Exercise. For each  $n \geq 2$  and  $i \in [n]$ ,

$$\text{cut} \left( \begin{array}{ccc} [n] & \longleftrightarrow & \{0 < \dots < i\} \\ \uparrow & & \uparrow \\ \{i < \dots < n\} & \longleftrightarrow & \{i\} \end{array} \right) \stackrel{\sim}{=} \begin{array}{ccc} \{1, \dots, n\}_+ & \xrightarrow{\text{collapse}} & \{1, \dots, i\}_+ \\ \downarrow \text{collapse} & & \downarrow \\ \{i+1, \dots, n\} & \longrightarrow & * \end{array}$$

Cor. Let  $\mathcal{C}$  be an  $\infty$ -category with finite products and  $X \in \text{CMon}(\mathcal{C})$ . Then the functor

$$\mathbb{A}^{\text{op}} \xrightarrow{\text{cut}} \text{Set}_*^{\text{fin}} \longrightarrow \mathcal{C}$$

is a monoid object. Hence precomposition with  $\text{cut}$  defines a forgetful functor

$$-\circ \text{cut}: \text{CMon}(\mathcal{C}) \longrightarrow \text{Mon}(\mathcal{C})$$

Def. A commutative monoid  $X$  is a group if the underlying monoid  $X \circ \text{cut} \in \text{Mon}(\mathcal{C})$  is a group.

Def. A Symmetric monoidal  $\infty$ -category is a commutative monoid in  $\text{Cat}_\infty$ .

Observe. If  $\mathcal{C}$  and  $\mathcal{D}$  are  $\infty$ -categories with finite products and  $F: \mathcal{C} \rightarrow \mathcal{D}$  preserves finite products, then postcomposition with  $F$  defines functors

$$\text{Mon}(\mathcal{C}) \rightarrow \text{Mon}(\mathcal{D}) \quad \text{and} \quad \text{CMon}(\mathcal{C}) \rightarrow \text{CMon}(\mathcal{D})$$

compatible with the forgetful functors.

Example. Recall that the inclusion  $\text{An} \hookrightarrow \text{Cat}_\infty$  preserves limits and admits a right adjoint  $\mathcal{C} \mapsto \mathcal{C}^\sim$ . Hence we obtain an adjunction

$$\text{CMon}(\text{An}) \rightleftarrows \text{CMon}(\text{Cat}_\infty)$$

That is, every symmetric monoidal  $\infty$ -category  $\mathcal{C}$  gives rise to a commutative monoid in  $\text{anima}$ .

Ex. Consider the category  $\text{Set}^{\text{fin}}$  with symmetric monoidal structure given by disjoint union. Then

$$\text{Set}^{\text{fin}}, \simeq \coprod_{n \geq 0} B\Sigma_n$$

is a commutative monoid in  $\text{anima}$ . One can show this is the free commutative monoid on a single generator.

## Group Completion

**Proposition** The inclusion  $\mathcal{C}\text{Grp}(\mathcal{A}n) \hookrightarrow \mathcal{C}\text{Mon}(\mathcal{A}n)$  admits a left adjoint  $X \mapsto X^{\text{gp}}$  called group completion.

**Recall.** By the Freudenthal suspension theorem, for a fixed  $k \geq 0$ , the maps

$$\pi_{n+k}(S^n) \longrightarrow \pi_{n+1+k}(S^{n+1})$$

are isomorphisms for  $n \geq k+2$ .

> The  $k$ -th stable homotopy group of spheres is

$$\pi_k(S) := \operatorname{colim}_{n \rightarrow \infty} \pi_{n+k}(S^n)$$

**Thm (Barratt-Priddy-Quillen).** For each  $n \geq 0$ , there is a canonical isomorphism

$$\pi_n((\text{Set}^{\text{fin}})^{\text{gp}}) \xrightarrow{\sim} \pi_n(S)$$

# Algebraic K-theory

Def. Let  $A$  be a ring. Regard the category  $\text{Proj}_A$  of finite projective  $A$ -modules as a symmetric monoidal category under direct sum. The connective algebraic K-theory of  $A$  is

$$K^{\text{cn}}(A) := (\text{Proj}_A^{\simeq})^{\text{gp}}.$$

> For  $n \geq 0$ , we write  $K_n(A) := \pi_n(K^{\text{cn}}(A))$ .

Thm (Quillen, 1972). Let  $q$  be a prime power. For  $i \geq 0$ , we have

$$K_n(\mathbb{F}_q) \simeq \begin{cases} \mathbb{Z}, & n = 0 \\ 0, & n > 0 \text{ even} \\ \mathbb{Z}/(q^i - 1), & n = 2i - 1 \end{cases}$$

> This is essentially the only complete computation of algebraic K-theory.

- > Though there have been remarkable recent advances by Antieau-Krause-Nikolaus using prismatic cohomology.
- > There are deep connections between K-theory and number theory.

Thm (Kunihara) TFAE:

(1) For  $i > 0$ , we have  $K_{4i}(\mathbb{Z}) = 0$

(2) Kummer-Vandiver Conjecture For each prime  $p$ , the number  $p$  does not divide the [class number] of the maximal real subfield of the cyclotomic field  $\mathbb{Q}(\zeta_p)$

size of class group

## Motivation for Spectra

- > Our goal was to understand the homotopy-theory refinement of abelian groups. In some sense, we've done that.
- > It is technically possible to formulate everything we need using  $\text{CGrp}(A_n)$ , but there is an important feature that  $\text{CGrp}(A_n)$  is missing.

Notation. Let  $A$  be a ring. Write

$$D(A) := \text{Ch}(A)[q, q^{-1}]$$

↳ all chain complexes

We call  $D(A)$  the derived  $\infty$ -category of  $A$ , and will later denote it simply by  $\text{Mod}_A$ .

Exercise. Let  $C_* \in D(A)$ . Convince yourself that the shifts

$$C_*[\pm 1] := (C_{*\pm 1}, d = (-1)^{\pm 1} d)$$

fit into squares

$$\begin{array}{ccc} C_*[-1] & \longrightarrow & 0 \\ \downarrow & & \downarrow \\ 0 & \longrightarrow & C_* \end{array} \quad \text{and} \quad \begin{array}{ccc} C_* & \longrightarrow & 0 \\ \downarrow & & \downarrow \\ 0 & \longrightarrow & C_*[1] \end{array}$$

that are both pullbacks and pushouts in  $D(A)$ . In particular, the functors  $(-)[\pm 1]: D(A) \rightarrow D(A)$  are inverse equivalences of  $\infty$ -categories.

> The  $\infty$ -category  $\mathrm{Grp}(An)$  does not satisfy this property:

Lemma. Let  $A$  be an abelian group in  $\text{Set}$ . Then the square in  $\text{CGrp}(A_n)$

$$\begin{array}{ccc} 0 & \longrightarrow & 0 \\ \downarrow & & \downarrow \\ 0 & \longrightarrow & A \end{array}$$

is a pullback.

Axiom. The inclusion  $\text{Set} \hookrightarrow A_n$  admits a left adjoint  $\pi_0: A_n \rightarrow \text{Set}$  that preserves products.

Proof of Lemma.

Note that since  $\text{Set} \hookrightarrow A_n$  is closed under limits,

$$\text{CGrp}(\text{Set}) \hookrightarrow \text{CGrp}(A_n)$$

is closed under limits. So we can compute the pullback in the usual category of abelian groups. This is clearly zero.  $\square$

Upshot  $\mathcal{C}\text{Grp}(\text{Ab})$  is really analogous to the connective part

$$D_{\geq 0}(\mathbb{Z}) = \text{Ch}_{\geq 0}(\mathbb{Z})[q, q^{-1}]$$

of the derived  $\infty$ -category of abelian groups.

> Spectra are the enlargement of  $\mathcal{C}\text{Grp}(\text{Ab})$  analogous to  $D(\mathbb{Z})$ .

# Spectra

> There are many ways of defining these. One way is to axiomatize reduced homology theories.

Def. An anima  $X$  is finite if  $X$  is in the smallest full subcategory of  $An$  containing  $\emptyset$  and  $*$  and closed under pushouts.

> Equivalently,  $X$  is the underlying anima of a finite CW complex.

We write  $An^{fin} \subset An$  for the full subcat. of finite anima.

Def. A spectrum is a functor  $E: An_*^{fin} \longrightarrow An$  satisfying

(1)  $E(*) \xrightarrow{\sim} *$

(2)  $E$  carries pushout squares to pullback squares.

We write  $Sp$  for the  $\infty$ -category of spectra and

$$\Omega^\infty: Sp \longrightarrow An$$

for the functor  $\Omega^\infty E := E(\{1\}_+)$  given by evaluating on the 0-sphere  $S^0 = \{1\}_+$ .

Obs. If  $X$  and  $Y$  are pointed anima, then

$$\begin{array}{ccc} X \vee Y & \xrightarrow{\text{Collapse}} & Y \\ \text{Collapse} \downarrow & & \downarrow \\ X & \longrightarrow & * \end{array}$$

is a pushout

> Taking  $X$  and  $Y$  to be finite pointed sets, we see that  $\Omega^\infty$  factors as

$$\begin{array}{ccccc} Sp & \xrightarrow{\text{restrict}} & CMon(A_n) & \xrightarrow{\text{forget}} & A_n \\ \cap & & \cap & & \\ \text{Fun}(A_n^{\text{fin}}_*, A_n) & & \text{Fun}(Set_*^{\text{fin}}, A_n) & & \end{array}$$

Exercise Convince yourself that  $\Omega^\infty Sp \rightarrow CMon(A_n)$  factors through  $Grp(A_n)$ .

Definition/Computation. Write  $S^0 = \{1\}$ . For each  $n \geq 1$ , there is a pushout square of (pointed) anima

$$\begin{array}{ccc} S^{n-1} & \longrightarrow & * \\ \downarrow & \lrcorner & \downarrow \\ * & \longrightarrow & S^n \end{array}$$

> More generally, if  $\mathcal{C}$  is an  $\infty$ -category with pushouts and a terminal object, the suspension of  $X \in \mathcal{C}$  is the pushout

$$\begin{array}{ccc} X & \longrightarrow & 1_{\mathcal{C}} \\ \downarrow & \lrcorner & \downarrow \\ 1_{\mathcal{C}} & \longrightarrow & \Sigma X \end{array}$$

Observe. Every Spectrum  $E$  comes equipped with equivalences

$$E(S^{n-1}) \xrightarrow{\sim} \Omega E(S^n) \quad \text{for } n \geq 1$$

> Often people write  $E_n = E(S^n)$

> There is an alternative perspective on Spectra, namely that they're determined by the anima  $(E_n)_{n \geq 0}$  and equivalences  $E_n \xrightarrow{\sim} \Omega E_{n+1}$ .

Prop. There is an equivalence of  $\infty$ -categories

$$\mathrm{Sp} \xrightarrow{\sim} \lim \left( \cdots \rightarrow \mathrm{An}_* \xrightarrow{\Omega} \mathrm{An}_* \xrightarrow{\Omega} \mathrm{An}_* \right)$$

given on objects by sending a spectrum  $E$  to the collection of anima  $(E_n)_{n \geq 0}$  equipped with the specified equivalences

$$E_n \xrightarrow{\sim} \Omega E_{n+1}$$

Remark The name 'spectrum' comes from this description of a spectrum as a sequence of anima  $(E_n)_{n \geq 0}$ .

> Probably not the best reason for the name...

## Connective Spectra & the recognition principle

Observe. Let  $E$  be a spectrum. For each  $i \geq 0$ , the equivs

$$E_n \xrightarrow{\sim} \Omega E_{n+1}$$

induce isomorphisms  $\pi_i(E_n) \xrightarrow{\sim} \pi_{i+1}(E_{n+1})$ .

Def. Let  $E$  be a spectrum and  $k \in \mathbb{Z}$ . The  $k$ -th homotopy group  $\pi_k(E)$  is the colimit

$$\pi_k(E) := \operatorname{colim}_{n \rightarrow \infty} \pi_{n+k}(E_n)$$

Here, the transition map  $\pi_{n+k}(E_n) \rightarrow \pi_{n+1+k}(E_{n+1})$  is the isomorphism explained above.

> So  $\pi_k(E) \cong \pi_{n+k}(E_n)$  for any  $n$  such that  $n+k \geq 0$ .

Def. A spectrum  $E$  is connective if for all  $k \leq 0$ , we have  $\pi_k(E) = 0$ .

> Equivalently, for each  $n \geq 0$ , the anima  $E_n$  only has nonzero homotopy in degrees  $\geq n$ .

We write  $Sp_{\geq 0} \subset Sp$  for the full subcategory spanned by the connective spectra.

Recognition principle (Segal). The functor  $\Omega^\infty Sp \rightarrow \mathcal{C}Grp(A_n)$  restricts to an equivalence

$$\Omega^\infty Sp_{\geq 0} \xrightarrow{\sim} \mathcal{C}Grp(A_n).$$

